Paper 2: knowledge-graph drug repurposing for liver fibrosis progression

Option 1 results (only drugs already used for a fibrosis-type liver disease are excluded). Knowledge graph only; 6,222 Paper 1 progression genes.

1. From all linked drugs to repurposing candidates

Step	Drugs	Genes
Drugs linked to our progression genes	3,027	1,072
Removed 620 drugs linked only through enzyme / transporter genes	2,407	958
Removed 1,015 early-stage drugs (below phase 2)	1,392	
Removed 59 withdrawn drugs	1,333	
Removed 64 drugs already used for a fibrosis-type liver disease = repurposing candidates	1,269	761
Drugs whose action on the gene is known (main list)	666	425
Drugs with at least one link opposite to the fibrosis direction	382	132
Stable drugs after final ranking	21	36

Fibrosis-type liver diseases: fatty liver, NASH, cirrhosis, fibrosis, portal hypertension and liver failure. Compared with the earlier rule (exclude any liver-disease use), 160 drugs come back, e.g. nintedanib, regorafenib and aspirin.

2. Final ranking

All 666 main-list drugs were scored with four equal-weight parts (target relevance, direction, specificity, graph support). The ranking was repeated **1,000 times** with slightly different weights. Drugs that stayed in the **top 25 in at least 80% of runs** are called stable: **21 drugs**.

3. Direction of the 21 stable drugs

The 21 drugs have 62 drug-target links. **27 links on 12 genes act opposite to the gene's fibrosis direction** (e.g. CXCR4 rises in fibrosis and mavorixafor blocks it; PPARA falls and fibrates activate it). **None act in the same direction.** The other 35 links, on 28 genes, have no up/down direction in the graph. Of the 12 genes, 4 are Tier 1 (CXCR4, BCL2, PIK3CG, CACNA1C); CACNA1C is in the 23-gene shortlist (barnidipine).

Table: the 21 stable drugs and the genes they act on

Rank	Drug	Acts opposite to fibrosis	Direction unknown
1	Mavorixafor	**CXCR4 [T1]**	CCR5
2	Drotaverine	PDE4A	**CACNA1C [T1] [Top23]**
3	Aplaviroc	CCR5	–
3	INCB-9471	CCR5	–
3	Maraviroc	CCR5	–
3	Vicriviroc	CCR5	–
7	Obatoclax	**BCL2 [T1]**	–
7	Venetoclax	**BCL2 [T1]**	–
9	Dexibuprofen	PTGS1, PTGS2	**BCL2 [T1]**, **THBD [T1]**, PLAT, PPARA, PPARG, SLC15A1
10	Fenoprofen	PTGS1, PTGS2	PPARA, PPARG
11	Duvelisib	**PIK3CG [T1]**, PIK3CD	–
11	TG100-115	**PIK3CG [T1]**, PIK3CD	–
13	Gemfibrozil	PPARA	SLCO1B1, SLCO1B3, SLCO2B1
14	Ibuprofen	PTGS1, PTGS2	**BCL2 [T1]**, **THBD [T1]**, PPARA, PPARG

Rank	Drug	Acts opposite to fibrosis	Direction unknown
15	Aspirin	PTGS1, PTGS2	**CCND1 [T1]**, **MYC [T1]**, CASP1, CASP3, EDNRA, HSPA5, MAPK4, MAPK6, MAPK7, PCNA, PRKAA1, PRKAA2, PRKAG1, PRKAG2, RPS6KA3
16	Pitavastatin	HMGCR	ITGAL
17	Lovastatin	HMGCR	HDAC2, ITGAL
18	Barnidipine	**CACNA1C [T1] [Top23]**	–
19	R-406	SYK	–
21	Ciprofibrate	PPARA	–
21	Clofibrate	PPARA	–

Only drug-target links are shown; enzyme and transporter links are left out. [T1] = Paper 1 Tier 1 gene (bold); [Top23] = 23-gene shortlist. All 27 opposite-acting links are recorded in both DrugBank and Open Targets (via the knowledge graph). The full list, including enzyme and transporter links, can be shared as an Excel file.

These are repurposing hypotheses to test, not proven results.